```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df=pd.read_csv("/content/boston.csv")
```

```
df
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296.0	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242.0	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242.0	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222.0	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222.0	18.7	396.90	5.33	36.2
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273.0	21.0	391.99	9.67	22.4
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273.0	21.0	396.90	9.08	20.6
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273.0	21.0	396.90	5.64	23.9
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273.0	21.0	393.45	6.48	22.0
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273.0	21.0	396.90	7.88	11.9

506 rows × 14 columns

```
x=df.drop("MEDV",axis=1)
y=df["MEDV"]
```

Start coding or generate with AI.

```
x=np.array(x)
y=np.array(y).reshape(-1,1)
```

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(
    x,y,test_size=0.3,random_state=42
)
```

```
from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
x_train=scaler.fit_transform(x_train)
x_test=scaler.transform(x_test)
```

```
from sklearn.linear_model import Ridge
```

```
from sklearn.linear_model import RidgeCV
alphas=np.logspace(-3,3,7)
ridge_cv=RidgeCV(alphas=alphas,cv=10)#k=10 cross vadlidation
ridge_cv.fit(x_train,y_train)
print("best alpha",ridge_cv.alpha_)
```

```
best alpha 10.0
```

```
best_ridge=Ridge(alpha=ridge_cv.alpha_)
best_ridge.fit(x_train,y_train)
y_pred=best_ridge.predict(x_test)
```

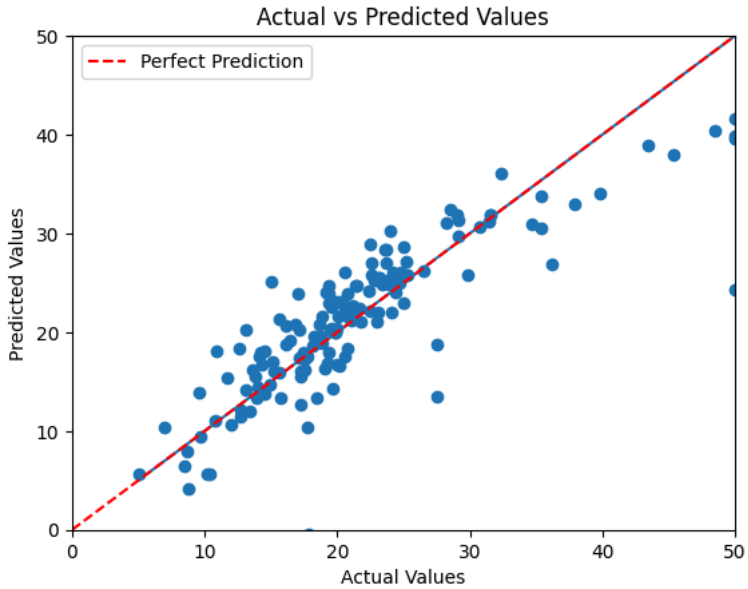
```
from sklearn.metrics import mean_squared_error,r2_score
```

```
mse = mean_squared_error(y_test, y_pred)
rmse=np.sqrt(mse)
r2=r2_score(y_test,y_pred)
```

```
print("Mean Squared Error:", mse)
print("Root Mean Squared Error:", rmse)
print("R-squared:", r2)
```

```
Mean Squared Error: 21.81124307149158
Root Mean Squared Error: 4.670250857447765
R-squared: 0.7072830902371281
```

```
import matplotlib.pyplot as plt
plt.scatter(y_test,y_pred)
plt.plot([min(y_test),max(y_test)],[min(y_test),max(y_test)])
max_val = max(np.max(y_test), np.max(y_pred))
plt.xlim(0, max_val)
plt.ylim(0, max_val)
plt.plot([0, max_val], [0, max_val], 'r--', label='Perfect Prediction')
plt.legend()
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted Values")
plt.show()
```




```
from sklearn.linear_model import Lasso
```

```
from sklearn.linear_model import LassoCV
alphas=np.logspace(-3,3,7)
lasso_cv=LassoCV(alphas=alphas,cv=10)
lasso_cv.fit(x_train,y_train)
print("best alpha",lasso_cv.alpha_)
```

```
best alpha 0.001
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_coordinate_descent.py:1664: DataConversionWarning: A column-ve
  y = column_or_1d(y, warn=True)
```

```
best_ridge=Lasso(alpha=lasso_cv.alpha_)
best_ridge.fit(x_train,y_train)
y_pred=best_ridge.predict(x_test)
```

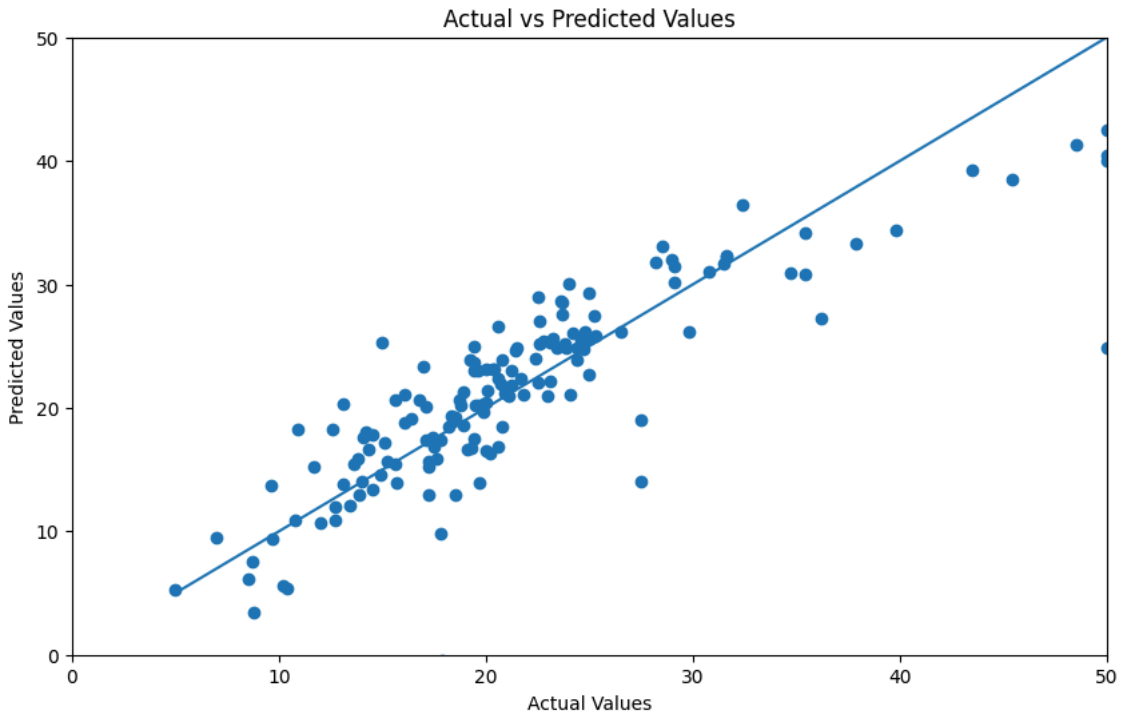
```
from sklearn.metrics import mean_squared_error,r2_score
```

```
mse = mean_squared_error(y_test, y_pred)
rmse=np.sqrt(mse)
r2=r2_score(y_test,y_pred)
print("Mean Squared Error:", mse)
print("Root Mean Squared Error:", rmse)
print("R-squared:", r2)
```

```
Mean Squared Error: 21.521510718492532
Root Mean Squared Error: 4.63912822828735
R-squared: 0.7111714316191495
```

```
plt.figure(figsize=(10,6))
plt.scatter(y_test,y_pred)
plt.plot([min(y_test),max(y_test)],[min(y_test),max(y_test)])
```

```
max_val=max(np.max(y_test),np.max(y_pred))
plt.xlim(0,max_val)
plt.ylim(0,max_val)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted Values")
plt.show()
```



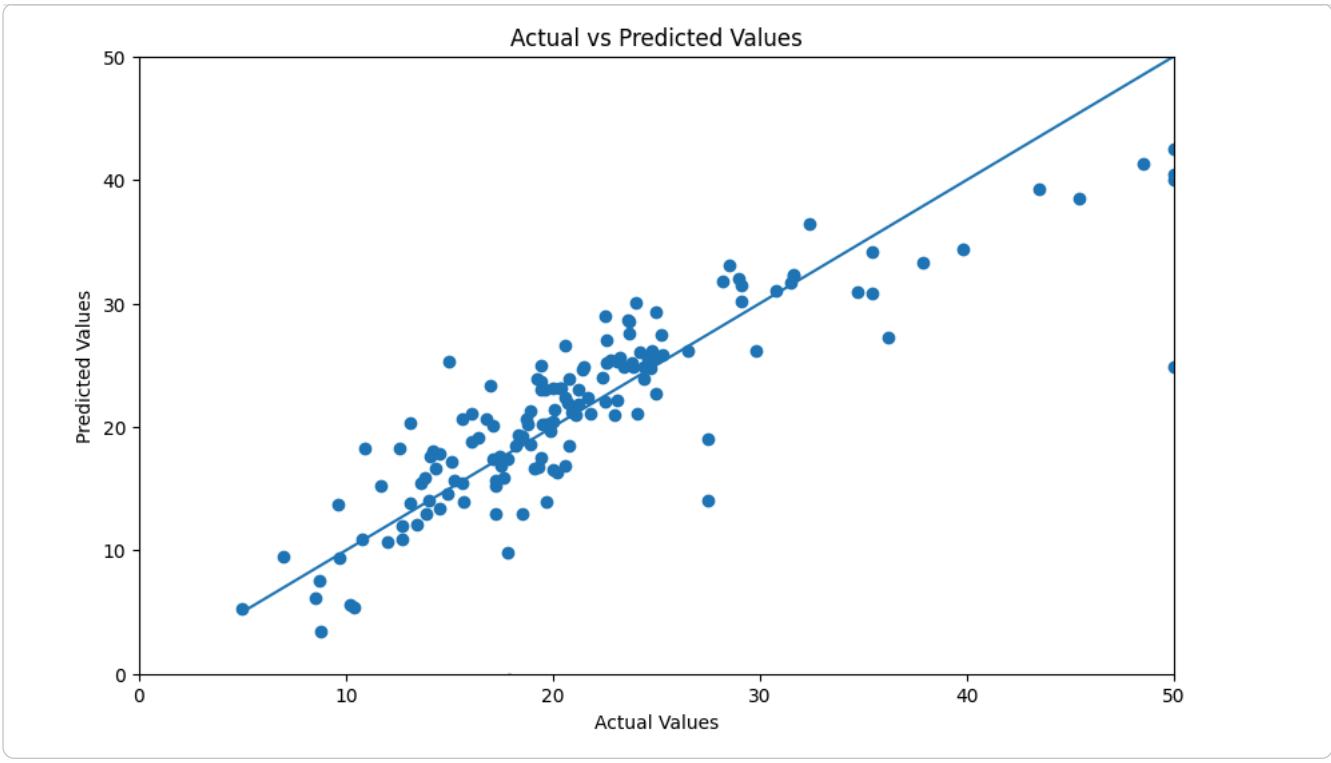

```
from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()
model.fit(x_train,y_train)
from sklearn.metrics import mean_squared_error,r2_score
```

```
mse=mean_squared_error(y_test,y_pred)
print("Mean squared error:",mse)
rmse=np.sqrt(mse)
print("Root mean squared error:",rmse)
R2= r2_score(y_test,y_pred)
print("R2 score:",R2)
```

```
Mean squared error: 21.521510718492532
Root mean squared error: 4.63912822828735
R2 score: 0.7111714316191495
```

```
plt.figure(figsize=(10,6))
plt.scatter(y_test,y_pred)
plt.plot([min(y_test),max(y_test)],[min(y_test),max(y_test)])
max_val=max(np.max(y_test),np.max(y_pred))
plt.xlim(0,max_val)
plt.ylim(0,max_val)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted Values")
plt.show()
```


Actual vs Predicted Values
Predicted Values
Actual Values
0
10
20
30
40
50